微分几何第二章习题及有关解答

2020 Autumn

1. 计算弧长用公式$s(t)=\int_a^t |r'(u)|du$, 计算曲率用第2题的公式.

(1).

$$\begin{aligned}\boldsymbol{r}(t) &= (t, at^2),\\ s(t) &= \int \sqrt{1+4a^2t^2}dt = \frac{t}{2}\sqrt{1+4a^2t^2} + \frac{\sinh^{-1}(2at)}{4a},\\ \kappa(t) &= \frac{2a}{(1+4a^2t^2)^{3/2}}.\end{aligned}$$

(2).

$$\begin{aligned}\boldsymbol{r}(t) &= (a\cos t, b\sin t),\\ s(t) &= \int \sqrt{a^2\sin^2 t + b^2\cos^2 t}dt, (\text{椭圆积分})\\ \kappa(t) &= \frac{ab}{(a^2\sin^2 t + b^2\cos^2 t)^{3/2}}.\end{aligned}$$

(3).

$$\begin{aligned}\boldsymbol{r}(t) &= (a\cosh t, b\sinh t),\\ s(t) &= \int \sqrt{a^2\sinh^2 t + b^2\cosh^2 t}dt, (\text{椭圆积分})\\ \kappa(t) &= \frac{-ab}{(a^2\sinh^2 t + b^2\cosh^2 t)^{3/2}}.\end{aligned}$$

(4).

$$\begin{aligned}\boldsymbol{r}(t) &= (t, a\cosh\frac{t}{a}),\\ s(t) &= \int \cosh\frac{t}{a}dt = a\sinh\frac{t}{a},\\ \kappa(t) &= \frac{1}{a\cosh^2(t/a)}.\end{aligned}$$

2.

$$
\begin{aligned}
ds &= |\boldsymbol{r}'(t)|dt = \sqrt{x'^2 + y'^2}dt, \\
\boldsymbol{t} &= \frac{\boldsymbol{r}'(t)}{|\boldsymbol{r}'(t)|} = \left(\frac{x'}{\sqrt{x'^2 + y'^2}}, \frac{y'}{\sqrt{x'^2 + y'^2}}\right), \\
\boldsymbol{n} &= \left(\frac{-y'}{\sqrt{x'^2 + y'^2}}, \frac{x'}{\sqrt{x'^2 + y'^2}}\right), \\
\kappa &= \frac{d\boldsymbol{t}}{ds} \cdot \boldsymbol{n} = \frac{d\boldsymbol{t}/dt}{ds/dt} \cdot \boldsymbol{n} = \frac{x'y'' - x''y'}{\left(x'^2 + y'^2\right)^{3/2}}.
\end{aligned}
$$

3. 对$\boldsymbol{r}(\theta) = (f(\theta)\cos\theta, f(\theta)\sin\theta)$用第2题的结果.

4. 计算曲率和挠率用第5题的公式.

(1).

$$
\kappa(t) = \frac{a\sqrt{a^2 + b^2\cosh 2t}}{\left(b^2 + a^2\cosh 2t\right)^{3/2}}, \quad \tau(t) = \frac{b}{a^2 + b^2\cosh 2t}.
$$

(2).

$$
\kappa(t) = \frac{3\sqrt{11}}{\left(9 + 22t^2\right)^{3/2}}, \quad \tau(t) = 0.
$$

(3).

$$
\kappa(t) = \frac{a}{a^2 + b^2}, \quad \tau(t) = \frac{-b}{a^2 + b^2}.
$$

(4).

$$
\kappa(t) = \tau(t) = \frac{\sqrt{2}t^2}{a(1 + t^2)^2}.
$$

5.

$$
\begin{aligned}
\boldsymbol{r}' &= \dot{\boldsymbol{r}}\frac{ds}{dt}, \\
\boldsymbol{r}'' &= \ddot{\boldsymbol{r}}(\frac{ds}{dt})^2 + \dot{\boldsymbol{r}}\frac{d^2s}{dt^2}, \\
\boldsymbol{r}''' &= \dddot{\boldsymbol{r}}(\frac{ds}{dt})^3 + 3\ddot{\boldsymbol{r}}\frac{ds}{dt}\frac{d^2s}{dt^2} + \dot{\boldsymbol{r}}\frac{d^3s}{dt^3},
\end{aligned}
$$

代入

$$
\begin{aligned}
\dot{\boldsymbol{r}} &= \boldsymbol{t}, \\
\ddot{\boldsymbol{r}} &= \kappa\boldsymbol{n}, \\
\dddot{\boldsymbol{r}} &= \dot{\kappa}\boldsymbol{n} + \kappa(-\kappa\boldsymbol{t} + \tau\boldsymbol{b}),
\end{aligned}
$$

得到

$$\boldsymbol{r}' \times \boldsymbol{r}'' = \dot{\boldsymbol{r}} \times \ddot{\boldsymbol{r}} (\frac{ds}{dt})^3 = \kappa |\boldsymbol{r}'|^3 \boldsymbol{b},$$
$$(\boldsymbol{r}', \boldsymbol{r}'', \boldsymbol{r}''') = (\dot{\boldsymbol{r}}, \ddot{\boldsymbol{r}}, \dddot{\boldsymbol{r}}) (\frac{ds}{dt})^6 = \kappa^2 |\boldsymbol{r}'|^6 \tau = |\boldsymbol{r}' \times \boldsymbol{r}''|^2 \tau.$$

6.

$$\begin{aligned}
\boldsymbol{r}'(s) &= (\frac{\sqrt{1+s}}{2}, -\frac{\sqrt{1-s}}{2}, \frac{1}{\sqrt{2}}), \quad |\boldsymbol{r}'(s)| = 1, \\
\kappa = \tau &= \frac{1}{2\sqrt{2}\sqrt{1-t^2}}, \\
\boldsymbol{t} = \boldsymbol{r}' &= \left(\frac{\sqrt{1+s}}{2}, -\frac{\sqrt{1-s}}{2}, \frac{1}{\sqrt{2}} \right), \\
\boldsymbol{n} = \frac{\boldsymbol{t}'}{\kappa} &= \left(\sqrt{\frac{1-s}{2}}, \sqrt{\frac{1+s}{2}}, 0 \right), \\
\boldsymbol{b} = \boldsymbol{t} \times \boldsymbol{n} &= \left(-\frac{\sqrt{1+s}}{2}, \frac{\sqrt{1-s}}{2}, \frac{1}{\sqrt{2}} \right).
\end{aligned}$$

7. 用到了光滑函数e^{-1/t^2}的性质, 不熟的翻一下微积分书.
(1). 三个分量均为C^∞函数, 且第二分量的导数恒不为零, 因此$\boldsymbol{r}(t)$是正则曲线. 在$t=0$处求导:

$$\boldsymbol{r}'(0) = (0,1,0), \quad \boldsymbol{r}''(0) = (0,0,0).$$

(2). 用第5题的公式算出$\kappa = \dfrac{2t^{-6}e^{-1/t^2}|2-3t^2|}{(1+4t^{-6}e^{-2/t^2})^{3/2}}$, 在$t=0, t^2=\dfrac{2}{3}$处曲率为零, Frenet标架不唯一.
在$(-\infty, -\sqrt{\dfrac{2}{3}})$上计算Frenet标架:

$$\begin{aligned}
\boldsymbol{t} &= \frac{\boldsymbol{r}'}{|\boldsymbol{r}'|} = \frac{(2t^{-3}e^{-1/t^2}, 1, 0)}{\sqrt{1+4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{n} &= \frac{\dot{\boldsymbol{t}}}{\kappa} = \frac{\boldsymbol{t}'}{\kappa|\boldsymbol{r}'|} = \frac{(-1, 2t^{-3}e^{-1/t^2}, 0)}{\sqrt{1+4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{b} &= \boldsymbol{t} \times \boldsymbol{n} = (0,0,1).
\end{aligned}$$

在$(-\sqrt{\dfrac{2}{3}}, 0)$上计算Frenet标架:

$$\begin{aligned}
\boldsymbol{t} &= \frac{\boldsymbol{r}'}{|\boldsymbol{r}'|} = \frac{(2t^{-3}e^{-1/t^2}, 1, 0)}{\sqrt{1+4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{n} &= \frac{\dot{\boldsymbol{t}}}{\kappa} = \frac{\boldsymbol{t}'}{\kappa|\boldsymbol{r}'|} = \frac{(1, -2t^{-3}e^{-1/t^2}, 0)}{\sqrt{1+4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{b} &= \boldsymbol{t} \times \boldsymbol{n} = (0,0,-1).
\end{aligned}$$

当$t \to 0^-$时Frenet标架的极限为$\{\boldsymbol{e}_2, \boldsymbol{e}_1, -\boldsymbol{e}_3\}$.

在$(0, \sqrt{\frac{2}{3}})$上计算Frenet标架:

$$\begin{aligned}
\boldsymbol{t} &= \frac{\boldsymbol{r}'}{|\boldsymbol{r}'|} = \frac{(0, 1, 2t^{-3}e^{-1/t^2})}{\sqrt{1 + 4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{n} &= \frac{\dot{\boldsymbol{t}}}{\kappa} = \frac{\boldsymbol{t}'}{\kappa|\boldsymbol{r}'|} = \frac{(0, -2t^{-3}e^{-1/t^2}, 1)}{\sqrt{1 + 4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{b} &= \boldsymbol{t} \times \boldsymbol{n} = (1, 0, 0).
\end{aligned}$$

当$t \to 0^+$时Frenet标架的极限为$\{\boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_1\}$.

在$(\sqrt{\frac{2}{3}}, +\infty)$上计算Frenet标架:

$$\begin{aligned}
\boldsymbol{t} &= \frac{\boldsymbol{r}'}{|\boldsymbol{r}'|} = \frac{(0, 1, 2t^{-3}e^{-1/t^2})}{\sqrt{1 + 4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{n} &= \frac{\dot{\boldsymbol{t}}}{\kappa} = \frac{\boldsymbol{t}'}{\kappa|\boldsymbol{r}'|} = \frac{(0, 2t^{-3}e^{-1/t^2}, -1)}{\sqrt{1 + 4t^{-6}e^{-2/t^2}}}, \\
\boldsymbol{b} &= \boldsymbol{t} \times \boldsymbol{n} = (-1, 0, 0).
\end{aligned}$$

8. $f(t) = |\boldsymbol{r}(t_0) - \boldsymbol{p}_0|^2$在$t = t_0$取最小值.

9. 注意: 我们简单地认为, 若曲率在一点不为零, 则曲率恒不为零.
(1). 条件为$\boldsymbol{r}(s) + \lambda(s)\boldsymbol{t}(s) = \boldsymbol{p}_0$, 其中$\lambda(s)$为$C^\infty$函数. 求导有

$$(1 + \lambda')\boldsymbol{t} + \lambda\kappa\boldsymbol{n} = 0,$$

于是

$$1 + \lambda' = 0, \quad \lambda\kappa = 0,$$

由此推出$\kappa = 0$.

(2). 条件为$\boldsymbol{r}(s) + \lambda(s)\boldsymbol{n}(s) = \boldsymbol{p}_0$, 其中$\lambda(s)$为$C^\infty$函数. 求导有

$$(1 - \lambda\kappa)\boldsymbol{t} + \lambda'\boldsymbol{n} + \lambda\tau\boldsymbol{b} = 0,$$

于是

$$1 - \lambda\kappa = 0, \quad \lambda' = 0, \quad \lambda\tau = 0,$$

由此推出λ为非零常数, 这说明κ为非零常数且$\tau = 0$.

10. 与p24性质4.1类似, 用到了p13第5题的结果.
反向刚体运动时$\boldsymbol{b}$会多出一个负号, 因此挠率会多出一个负号.

11.

$$\begin{aligned}
\tilde{\boldsymbol{t}} &= \tilde{\boldsymbol{r}}' = \boldsymbol{b}, \\
\tilde{\boldsymbol{t}}' &= \tilde{\kappa}\tilde{\boldsymbol{n}} = \boldsymbol{b}' = -\tau\boldsymbol{n}, \\
\Rightarrow \tilde{\kappa} &= \tau, \tilde{\boldsymbol{n}} = -\boldsymbol{n}, \\
\tilde{\boldsymbol{b}} &= \tilde{\boldsymbol{t}} \times \tilde{\boldsymbol{n}} = -\boldsymbol{b} \times \boldsymbol{n} = \boldsymbol{t}, \\
\tilde{\boldsymbol{b}}' &= -\tilde{\tau}\tilde{\boldsymbol{n}} = \boldsymbol{t}' = \kappa\boldsymbol{n}, \\
\Rightarrow \tilde{\tau} &= \kappa.
\end{aligned}$$

12. 注意: s不一定是$\tilde{\boldsymbol{r}}$的弧长参数.

$$\begin{aligned}
\tilde{\boldsymbol{r}}' &= \kappa\boldsymbol{n}, \\
\tilde{\boldsymbol{r}}'' &= -\kappa^2\boldsymbol{t} + \kappa'\boldsymbol{n} + \kappa\tau\boldsymbol{b}, \\
\tilde{\boldsymbol{r}}''' &= (-3\kappa\kappa')\boldsymbol{t} + [\kappa'' - \kappa(\kappa^2+\tau^2)]\boldsymbol{n} + [\kappa'\tau + (\kappa\tau)']\boldsymbol{b}, \\
\tilde{\boldsymbol{r}}' \times \tilde{\boldsymbol{r}}'' &= \kappa^2(\tau\boldsymbol{t} + \kappa\boldsymbol{b}), \\
(\tilde{\boldsymbol{r}}', \tilde{\boldsymbol{r}}'', \tilde{\boldsymbol{r}}''') &= \kappa^3(\kappa\tau' - \kappa'\tau), \\
\tilde{\kappa} &= \frac{\sqrt{\kappa^2+\tau^2}}{\kappa}, \\
\tilde{\tau} &= \frac{\kappa\tau' - \kappa'\tau}{\kappa(\kappa^2+\tau^2)}.
\end{aligned}$$

16. 由Taylor展开式$\boldsymbol{r}(s) = \boldsymbol{r}(0) + s\boldsymbol{t}(0) + \frac{s^2}{2}\kappa(0)\boldsymbol{n}(0) + o(s^2)$得到

$$\begin{aligned}
d^2(P,l) &= |(\boldsymbol{r}(s) - \boldsymbol{r}(0)) \times \boldsymbol{t}(0)|^2 = |\frac{s^2}{2}\kappa(0)\boldsymbol{b}(0) + o(s^2)|^2 \sim |\frac{s^2}{2}\kappa(0)|^2, \\
d^2(P_0,P) &= |\boldsymbol{r}(s) - \boldsymbol{r}(0)|^2 = |s\boldsymbol{t}(0) + o(s)|^2 \sim s^2.
\end{aligned}$$

17.

18. 注意: 求一次导就出一个负号.

(1). 用第2题的公式, $\tilde{\kappa}(t) = -\kappa(-t)$.

(2). 用第5题的公式, $\tilde{\kappa}(t) = \kappa(-t), \tilde{\tau}(t) = \tau(-t)$.

注: 这说明在反向参数变换下, 平面曲线的曲率会改变符号, 而空间曲线的曲率和挠率都不改变符号.

19. 设$\boldsymbol{v} = \lambda\boldsymbol{t} + \mu\boldsymbol{n} + \nu\boldsymbol{b}$, 代入条件有

$$\begin{aligned}
\boldsymbol{v} \times \boldsymbol{t} &= \nu\boldsymbol{n} - \mu\boldsymbol{b} = \boldsymbol{t}' = \kappa\boldsymbol{n}, \\
\boldsymbol{v} \times \boldsymbol{n} &= -\nu\boldsymbol{t} + \lambda\boldsymbol{b} = \boldsymbol{n}' = -\kappa\boldsymbol{t} + \tau\boldsymbol{b}, \\
\boldsymbol{v} \times \boldsymbol{b} &= \mu\boldsymbol{t} - \lambda\boldsymbol{n} = \boldsymbol{b}' = -\tau\boldsymbol{n},
\end{aligned}$$

于是$\lambda=\tau,\mu=0,\nu=\kappa$恰好满足要求.

20. 算出曲率$\kappa=\tilde{\kappa}=\dfrac{1}{4}$, 挠率$\tau=\tilde{\tau}=-\dfrac{1}{4}$, 用基本定理.

Solution to T22 in 2nd version

Without loss of generality, we may assume that t is the arc-length parameter.

(1). From equations of Frenet frame, we have

$$\boldsymbol{\alpha}' = \mathbf{t} + \frac{-\kappa \mathbf{t}}{\kappa} + \frac{\kappa'}{\kappa^2}\mathbf{n} = \frac{\kappa'}{\kappa^2}\mathbf{n},$$

which is vertical to $\mathbf{r}'$.

(2). Assume that

$$\boldsymbol{\alpha}(t, t') = \mathbf{r}(t') + \lambda(t, t')\mathbf{n}(t'),$$

then by choosing a small neighbourhood, we have that $\lambda(t, t')$ is a smooth and bounded function of (t, t').

$\boldsymbol{\alpha}(t, t')$ is on the normal line at $\mathbf{r}(t)$, so

$$\begin{aligned}
0 =& (\boldsymbol{\alpha}(t, t') - \mathbf{r}(t)) \cdot \mathbf{r}'(t) \\
=& (\mathbf{r}(t') - \mathbf{r}(t)) \cdot \mathbf{r}'(t) + \lambda(t, t')\mathbf{n}(t') \cdot \mathbf{r}'(t) \\
=& (\mathbf{r}(t') - \mathbf{r}(t)) \cdot \mathbf{r}'(t) + \lambda(t, t')(\mathbf{n}(t') - \mathbf{n}(t)) \cdot \mathbf{r}'(t) \\
=& |\mathbf{r}'(t)|^2(t' - t) + \lambda(t, t')(-\kappa(t))|\mathbf{r}'(t)|^2(t' - t) + o(|t - t'|).
\end{aligned}$$

Therefore, the limit $\lim_{t' \to t} \lambda(t, t')$ exists and equals to $\frac{1}{\kappa}$, i.e. $\lim_{t' \to t} \boldsymbol{\alpha}(t, t') = \boldsymbol{\alpha}(t)$.

Supplement

Exercise 1 *Let $C : \overrightarrow{r} = \overrightarrow{r(s)}$ is a smooth curve, neither a line nor a helix, parametrized by arc length s. We can show that the normal vector of C intersects with some vector in a fixed angle if and only if the curvature κ and the torsion τ satisfy the equation*

$$\tau + \frac{d}{ds}\left(\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}\right) = 0.$$

Proof.

($\Longrightarrow$) Let $\left\{\overrightarrow{T}, \overrightarrow{N}, \overrightarrow{B}\right\}$ be the Frenét frame along C. The hypothesis gives us that

$$\overrightarrow{N} \cdot \overrightarrow{a} = c \tag{1}$$

for some unit vector $\overrightarrow{a}$ and some constant c. Differentiating both sides, we get

$$\overrightarrow{T} \cdot \overrightarrow{a} = \frac{\tau}{\kappa}\overrightarrow{B} \cdot \overrightarrow{a}; \tag{2}$$

then, differentiating both sides again, we obtain

$$\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}\overrightarrow{N} \cdot \overrightarrow{a} = \overrightarrow{B} \cdot \overrightarrow{a}.$$

So we have, by (1),

$$\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}c = \overrightarrow{B} \cdot \overrightarrow{a}; \tag{3}$$

hence, doing the differentiation once again, it gives us

$$c\left[\tau + \frac{d}{ds}\left(\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}\right)\right] = 0.$$

It is easy to note that $c \neq 0$ (for exercise!). The necessary part is done.

($\Longleftarrow$) The equations (1), (2) and (3) inspire us to set a vector $\overrightarrow{A(s)}$ to be

$$\overrightarrow{A(s)} = \frac{\tau}{\kappa}\left(\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}\right)\overrightarrow{T} + \overrightarrow{N} + \left(\frac{\kappa^2+\tau^2}{\kappa\frac{d}{ds}\left(\frac{\tau}{\kappa}\right)}\right)\overrightarrow{B}.$$

If we could show $\overrightarrow{A(s)}$ is a constant vector, then

$$\overrightarrow{a} = \frac{\overrightarrow{A(s)}}{\left\|\overrightarrow{A(s)}\right\|}$$

will satisfies $\overrightarrow{N} \cdot \overrightarrow{a} =$ constant. So the sufficient part is completed. However, it is just a little bit tedious calculation!! ■